National University of Computer and Emerging Sciences

Lab Manual # 04
Object Oriented Programming
(CL2004)

Course Instructor	Usama Hassan
Lab Instructor (s)	Fareeha Ashfaq Marwa Khan
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Department of Computer Science
FAST-NU, Lahore, Pakistan

Objectives
After performing this lab, students shall be able to:
- ✓ Class declaration
- ✓ Data members and member functions
- ✓ Setter and getter functions
- ✓ Default and parameterized constructors

TASK 1:
Define a class student with the following specifications:

Data Members:

- Name
- Eng, Math, Science
- Total

Public Members:

- ***Take_data( )*** - A function to accept values for Ali, Ahmed, eng, science and invoke ***ctotal( )*** to calculate total.
- ***Show_data( )*** - A function to display all the data members on the screen.
- ***ctotal( )*** - A function to calculate Eng + Math + Science with float return type.

TASK 2:
A phone number, such as (212) 767-8900, can be thought of as having three parts: The Area code (212), the exchange (767), and the number (8900). Write a program that uses a ***CLASS*** to store these three parts of a phone number separately. Call the class Phone. Create two class variables of type phone. Initialize one, and have the user Input a number for the other one. Then display both numbers. The interchange might Look like this:

For Example:

INPUT:

Enter your area code, exchange, and number: 415 555 1212

OUTPUT:

My number is (212) 767-8900

Your number is (415) 555-1212

TASK 3:
Exercise 1:
- Create a class Date having following private data members:
  Int Day
  Int Month
  Int Year

- Create an object of Date “date1” and run your program

Exercise 2 [Default Constructor]:
- Write a default Constructor of Date that initializes the object to 1st January 1926 and prints “Default Constructor Called” in start.
- Now run your program and test what does date1 prints?

Exercise 3 [Print Function]:
- Implement a function Print in Date class which prints a date in following format: dd/mm/yyyy (e.g. 1/1/1926 for date1)
- Print object date1 in your main function and run the program.
- What does it print and how can we initialize the data of date1 at the time of creation?

Exercise 4 [Input Function]:
- Write a function Input in your Date class that takes input from user to populate a Date object.
- Call “date1.Input()” and “date1.Print()” in your driver program and test it.

Exercise 5 [Setters]:
- Create an object xmasDay using default constructor.
- Print xmasDay and see what it prints.
- Write Setters i.e. SetDay, SetMonth and SetYear in your class.
- Now set xmasDay to 25/12/2020 using Setters in main.

Exercise 6 [Getters]:
- Write Getters i.e. GetDay, GetMonth and GetYear in your date class.

Now print xmasDay using Getters in your Driver program.

TASK 4:

Consider the definition of the following class:

```
class Sample{
private:
      int x;
      double y;
public:
      Sample( );    //Constructor 1
      Sample(int); //Constructor 2
      Sample(int , int); //Constructor 3
      Sample(int, double); //Constructor 4
```

```
};
```

a) Write the definition of constructor 1 so that the private data members are initialized to 0.

b) Write the definition of constructor 2 so that the private data member x is initialized according to the value of the parameter, and the private data member y is initialized to 0.

c) Write the definition of constructor 3 and 4 so that the private data members are initialized to according to the values of parameters.